

Session 07 Homework — Extend Projects and Workspaces

CORE, STRETCH, and CHALLENGE are all required — every one of them must be completed. The three names describe increasing difficulty only, nothing about which parts you submit.

Context

This extends B4F Board's `api/` exactly where class left off: `ProjectsModule` and `WorkspacesModule` both exist, each with a service (`findAll` / `findOne` / `update`) and a controller exposing `GET` / `PATCH` routes, registered in `AppModule` .

Goal

Add one new lookup endpoint to Projects, one computed endpoint to Projects, and one new lookup endpoint to Workspaces — using only the Nest building blocks already taught in class.

What to build

`GET /projects/key/:key` (CORE), `GET /projects/stats` (STRETCH), and `GET /workspaces/slug/:slug` (CHALLENGE) — all three added to the existing services/controllers you already built in class.

CORE (required)

Add `GET /projects/key/:key` :

- `404` if no project has that key.
- Otherwise, return the matching project.

STRETCH (required)

Add `GET /projects/stats` , returning an object shaped like `{ total: number }` — the total number of projects.

Think carefully about **where** in the controller this route goes relative to the `:id` route you already have, before testing. If you get a surprising result, that placement is the first thing to check.

CHALLENGE (required)

Add `GET /workspaces/slug/:slug` :

- `404` if no workspace has that slug.
- Otherwise, return the matching workspace.

Rules

Allowed: everything taught in Session 07 — `@Controller`, `@Get`, `@Param`, `@Injectable`, constructor injection, and plain TypeScript/JavaScript logic inside a service method.

Not yet: no DTOs, no `class-validator` decorators, no `ValidationPipe`, no guards, no interceptors, no custom exception filters, no database of any kind, no new npm package. All validation is a plain `if` statement throwing a built-in Nest exception, exactly like `findOne` already does in class.

Test it

Use `curl` or Postman against your own running server (`npm run start:dev`). For each endpoint, test both a success case and the `404` case.

Done when

- All three endpoints exist and behave exactly as described above.
- `npm run start:dev` starts with no errors and no warnings in the terminal.
- You can explain, out loud, why your `stats` route's position in the file does or does not matter.

Working time

Expect roughly 30-40 minutes total across all three tiers.